Step 1: Prepare a Repository Template - Expanded Guide

What is a Template Repository?

A template repository serves as the foundation for your assignment. It contains all the materials students need to complete the assignment, structured in a way that's easy to understand and navigate.

Detailed Steps:

1. Create a New Repository

1. Go to GitHub.com and sign in to your account
2. Click the "+" icon in the upper right corner and select "New repository"
3. Name your repository (e.g., "r-basic-data-analysis-template")
4. Add a brief description (e.g., "Template for basic data analysis in R assignment")
5. Choose "Public" or "Private" (Public is fine for a template)
6. Check the box that says "Initialize this repository with a README"
7. Click "Create repository"

2. Set as Template Repository

1. Navigate to your new repository
2. Click on "Settings" in the top navigation bar
3. Scroll down to the "Template repository" section
4. Check the box labeled "Template repository"
5. This allows GitHub Classroom to create copies of this repository for each student

3. Structure Your Repository

1. Clone the repository to your local machine using:

   ```
   git clone https://github.com/yourusername/r-basic-data-analysis-template.git
   ```

2. Create a logical folder structure, for example:

   - `/data` - For datasets
   - `/instructions` - For detailed assignment instructions
   - `/starter-code` - For code templates students will build upon
   - `/tests` - For any test scripts to validate solutions

4. Add Assignment Materials

1. Create a comprehensive README.md file:

   - Assignment title and learning objectives

- Clear step-by-step instructions
- Due date and submission guidelines
- Grading criteria
- Resources for students

2. Add the dataset (e.g., student_scores.csv) to the `/data` folder
3. Create starter code files in the `/starter-code` folder:
   - analysis.R with code skeleton and TODOs
   - Include helpful comments to guide students
4. Add any additional resources:
   - Reference documents
   - Helpful examples
   - Links to relevant documentation

5. Include Optional Test Suite

1. If you want to include automated tests:
   - Add test files that check if student solutions meet requirements
   - For R, consider using the testthat package
   - Include instructions on how to run tests

6. Commit and Push Your Changes

1. Once you've added all materials, commit your changes:

```
git add .
git commit -m "Initial template setup with assignment materials"
git push origin main
```

7. Review the Repository

1. Go to your repository on GitHub.com
2. Make sure all files are correctly uploaded
3. Check that the README displays properly
4. Verify links work and files are accessible
5. Look at your repository as if you were a student to ensure clarity

8. Test the Template

1. Create a test copy of your template by clicking "Use this template" button
2. Go through the assignment as a student would
3. Make sure all instructions are clear and materials work properly
4. Make any necessary adjustments to the original template

Once your template repository is ready, you can move on to creating the assignment in GitHub Classroom as outlined in Step 2.

I'll provide you with step-by-step instructions for creating and distributing an assignment through GitHub Classroom, along with a simple R coding assignment example.

Creating and Distributing an Assignment in GitHub Classroom

Step 1: Prepare a Repository Template

1. Create a new repository on GitHub that will serve as your template
2. Add the assignment materials (instructions, starter code, etc.)
3. Include a README.md with clear instructions for students

Step 2: Create the Assignment in GitHub Classroom

1. Log in to your GitHub Classroom
2. Click on the classroom you've already created
3. Click the "Create assignment" button
4. Fill in the assignment details:
    - Assignment title (e.g., "Basic Data Analysis in R")
    - Deadline (optional)
    - Select "Individual assignment" or "Group assignment"
5. Enable the assignment repository to be private
6. Under "Add a template repository," select the repository you created in Step 1
7. Configure any additional settings (automatic grading, etc.)
8. Click "Create assignment"

Step 3: Distribute the Assignment

1. After creating the assignment, you'll see an invitation link
2. Share this link with your students via your course management system or email
3. When students click the link, they'll be prompted to accept the assignment
4. GitHub will automatically create a personal copy of the repository for each student

Step 4: Monitor Student Progress

1. From your GitHub Classroom dashboard, click on the assignment
2. You'll see a list of students who have accepted the assignment
3. You can view their repositories and track their progress

Example R Coding Assignment

Here's a simple R assignment example that you could use as a template:

Now you have both the assignment instructions and a sample dataset for your GitHub Classroom assignment. Here are a few additional tips for a successful implementation:

1. **Test your template repository before creating the assignment** to make sure everything works correctly
2. **Include clear assessment criteria** in your README so students understand how they'll be evaluated
3. **Consider adding tests** using the testthat package in R to help students verify their solutions
4. **Set up automatic grading** if desired, using GitHub Classroom's built-in features
5. **Be available for questions** as students might need help with both the R assignment and GitHub workflow
6. **Remind students to commit and push frequently** so they don't lose their work and you can see their progress

Would you like me to elaborate on any particular aspect of creating or managing GitHub Classroom assignments?